Interpretable artificial intelligence (TME286)

Lecture 3, 20260203

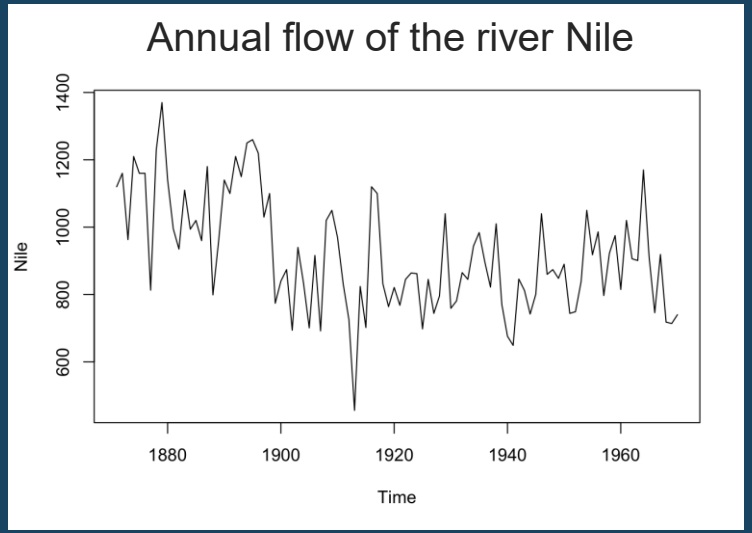
Time series prediction

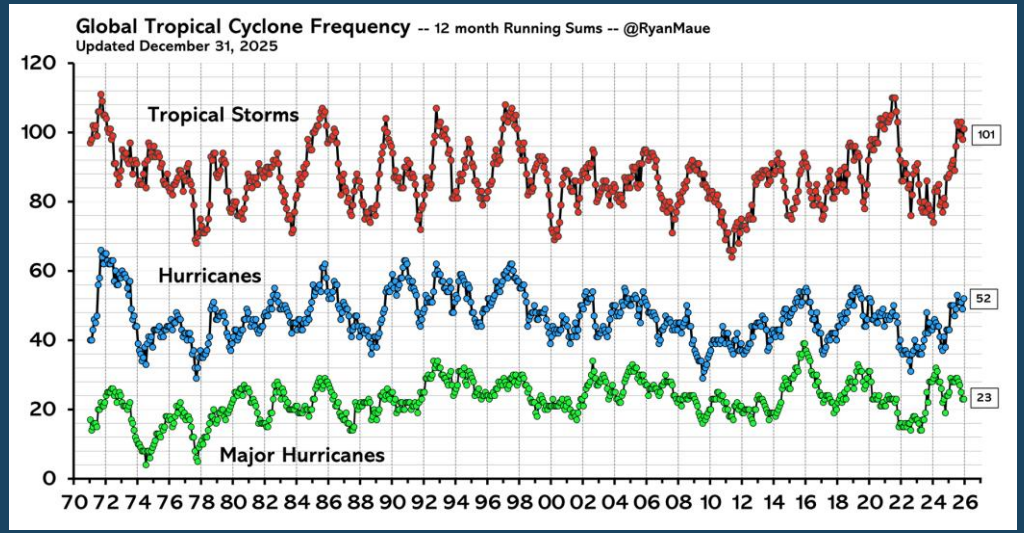
Today’s learning goals

- After this lecture you should be able to
  - Describe and discuss different types of time series
  - Describe and compare different linear models
  - Apply linear models to a time series prediction task
  - Describe and discuss the Black swan phenomenon
  - Apply LSTM networks to a time series prediction task
  - Briefly describe event prediction (as a classification task)

Time series (types)

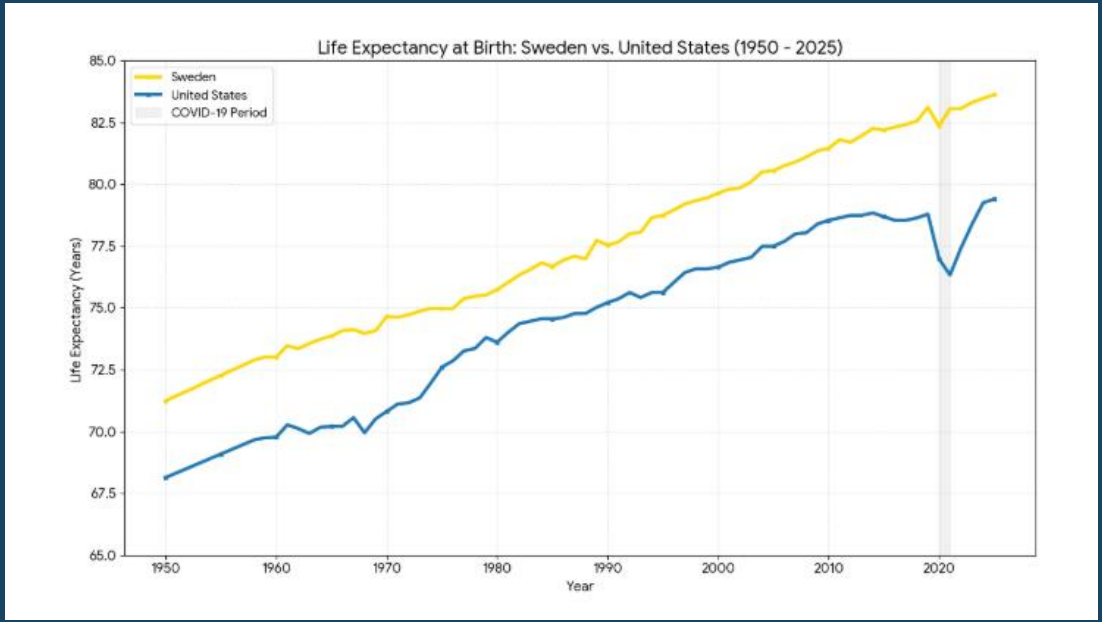
- Without trend:





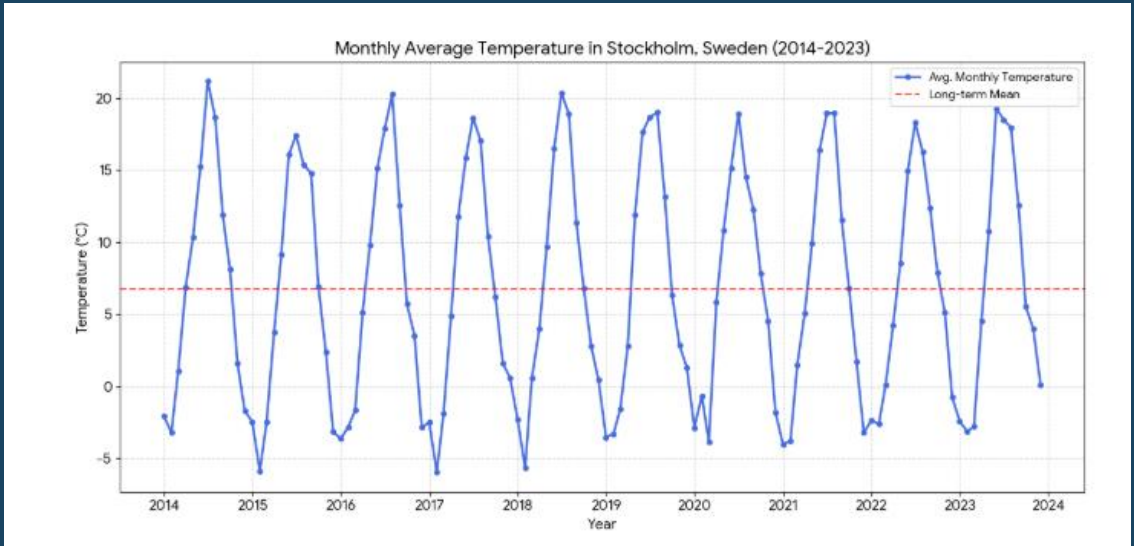
Time series (types)

- With trend:



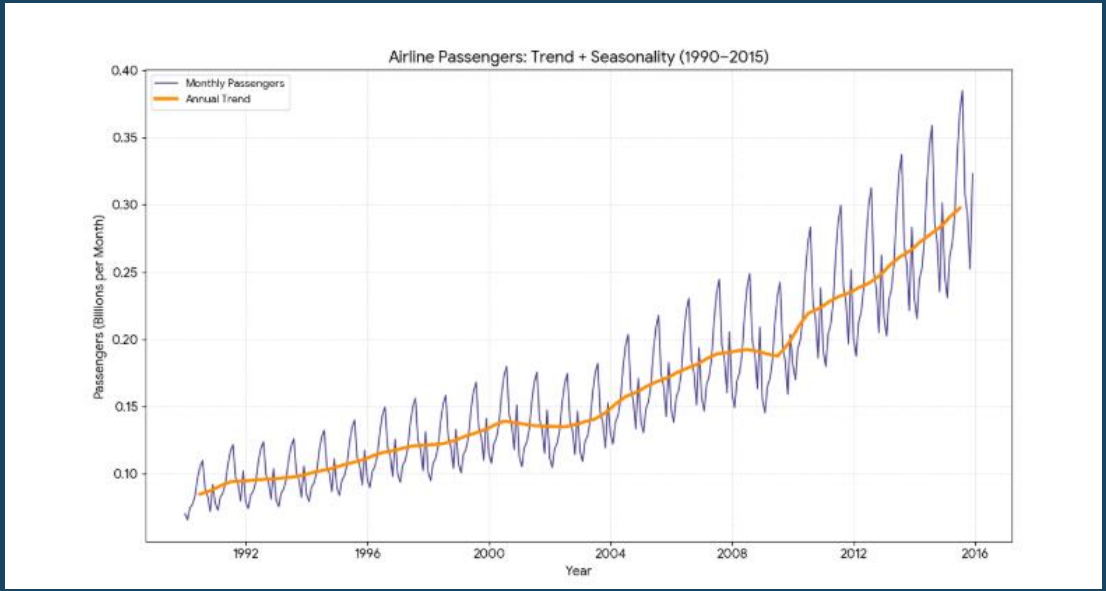
Time series (types)

- Seasonality, no trend:



Time series (types)

- Seasonality and trend:



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Linear models

- As in many other AI subfields, it is always worth considering linear models first, before trying more complex models.
- There are different kinds of linear model for time series predictions, namely ...
  - ... Autoregressive (AR) models
  - ... Moving average (MA) models
  - ... ARMA models (combining AR and MA)
  - ... ARIMA models (like ARMA, but handling data with trends)
- One can analyse a time series either in the time-domain or in the frequency-domain.
- Here we will focus on the time domain (more specifically: Univariate, discrete time series).

Linear time series prediction: AR(p)

Autoregressive (AR(p)) model:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t$$

- X_t: The value at the current time.
- c: A constant (intercept).
- $\phi_1, \ldots, \phi_p$: The coefficients (weights) the model learns.
- $X_{t-1}, \ldots, X_{t-p}$: The "lags" (past values).
- ϵ_t: White noise (the part of today's value that couldn't be predicted from the past).

Linear time series prediction: AR(p)

- The model can be trained with linear regression, but one must then first reformat the data (lagging it):
- Example (with p=2): Time series = {10,12,15,14,20}
- Lagged data for this simple example:

Target (X_t)	X_{t-1}	X_{t-2}
15	12	10
14	15	12
20	14	15

- There are better methods for training AR models though (described in Lecture 5).

Linear time series prediction: AR(p)

Moving average (MA(p)) model:

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q}$$

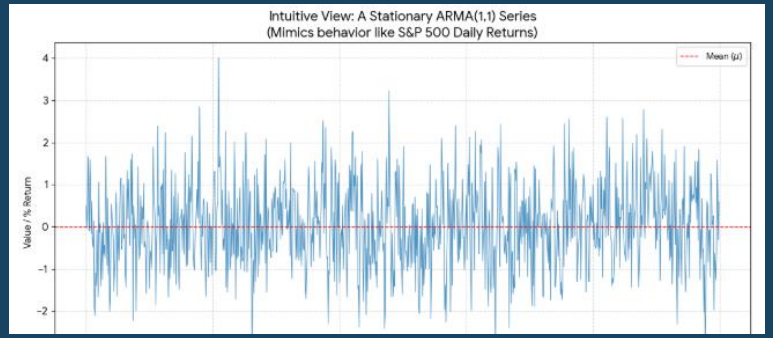
- X_t: The value at the current time.
- μ: The mean of the series (unlike AR, where c is just an intercept).
- ϵ_t: The current "shock" or white noise (error). It is assumed to be $N(0, \sigma^2)$.
- $\theta_1, \ldots, \theta_q$: The coefficients (weights) for the past shocks.
- $\epsilon_{t-1}, \ldots, \epsilon_{t-q}$: The lagged forecast errors.

Linear time series prediction: ARMA(p,q)

Moving average (ARMA(p,q)) model:

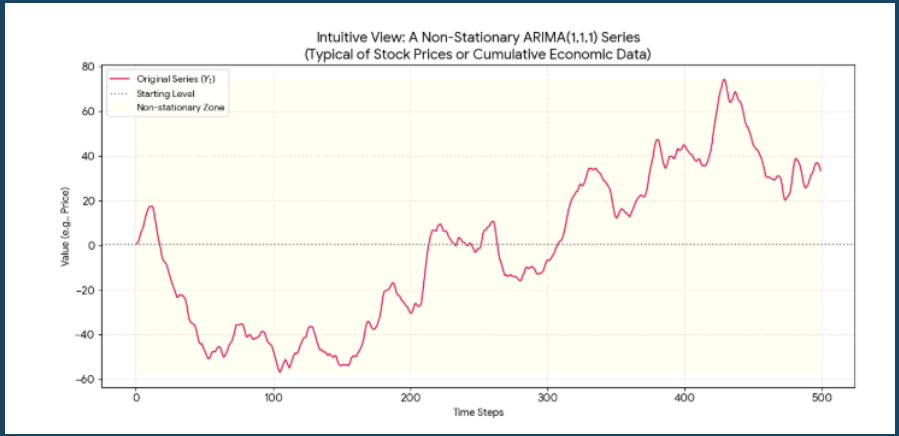
$$X_t = c + \underbrace{\phi_1 X_{t-1} + \cdots + \phi_p X_{t-p}}_{\text{AR Part}} + \underbrace{\theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}}_{\text{MA Part}} + \epsilon_t$$

MA, AR, and ARMA are applicable to stationary time series, without a trend:



Linear time series prediction: ARIMA(p,d,q)

- In an ARIMA(p,d,q) model one considers data differenced data, e.g., $X_t - X_{t-1}$
- Applying an ARIMA model to non-differenced data is the same as applying an ARMA model after first doing the differencing (manually).



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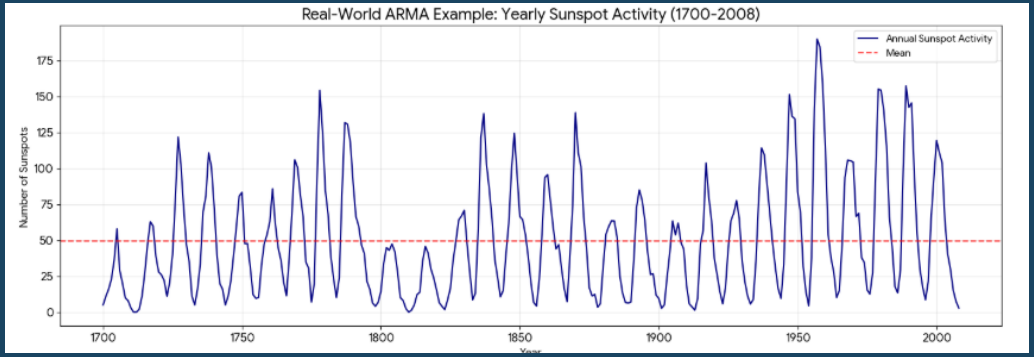
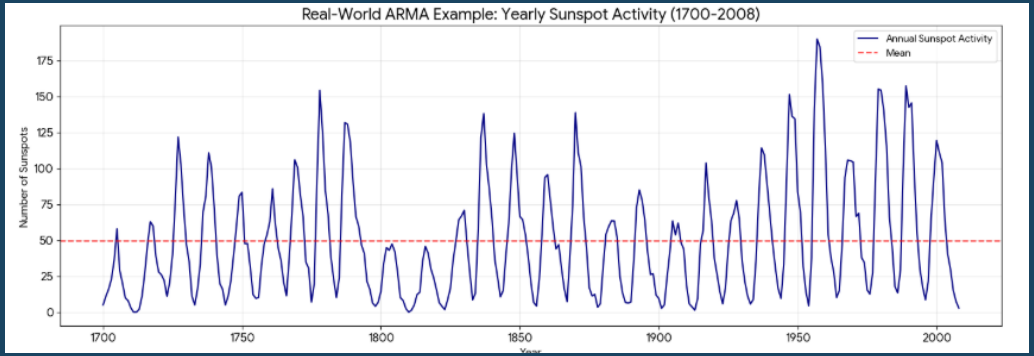
Applying linear models

- Three main steps
  - Model selection (*Is an ARMA model suitable for the case under study?*)
  - Parameter selection *(Which values of p and q should be used if the selected model is ARMA?)*
  - Model fitting *(Finding the values of the coefficients in the model)*



Applying linear models : Model selection

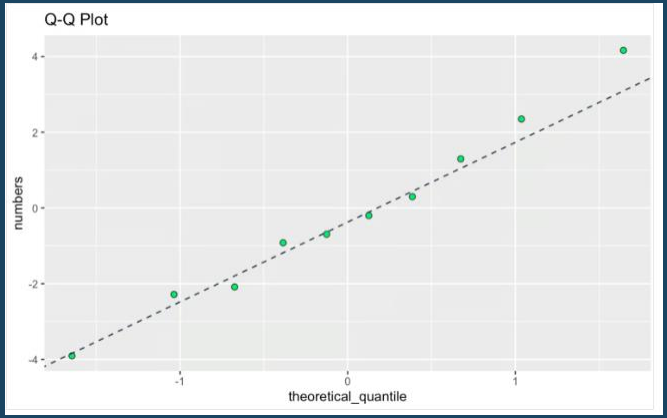
- Considerations:
  - Are the data stationary? (=> ARMA)
  - Is there a trend? (=> ARIMA, but de-trend by looking at differences => ARMA)
  - Is there a seasonal pattern? (=> Seasonal AR(I)MA, but seasonality can be removed => ARMA)



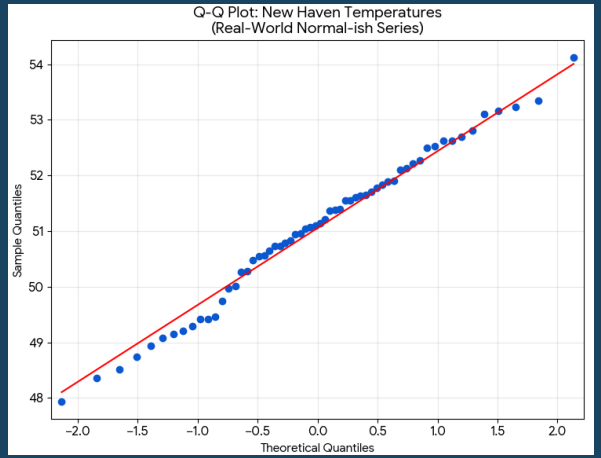
Testing for linearity: Q-Q-plot

- Here, we already assumed that a linear model is appropriate.
- One can *test* for that as well, however, for example, by checking whether or not the data are normal-distributed.
- Example: Q-Q-plotting
  - Sort the data from smallest to largest
  - Calculate cumulative probabilities P_i
  - Look up the value from the Normal distribution, $Y = \Phi^{-1}(P_i)$
  - Plot the data vs. Y
  - Look for deviations from linearity.

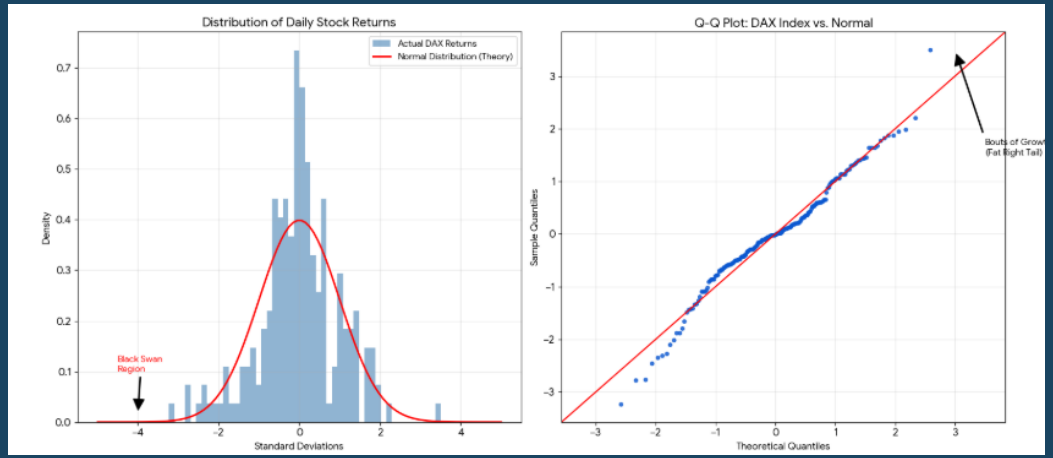
Numbers	Rank	Probability (prob)	Theoretical Quantile
-3.905363	1	0.05	-1.644854
-2.282611	2	0.15	-1.036433
-2.085952	3	0.25	-0.674490
-0.919770	4	0.35	-0.385321
-0.693255	5	0.45	-0.125661
-0.200144	6	0.55	0.125661
0.299856	7	0.65	0.385321
1.297350	8	0.75	0.674490
2.349959	9	0.85	1.036433
4.163738	10	0.95	1.644854



Testing for linearity: Q-Q-plot



Temperature data (~normal-distributed)



Stock returns (with black swans!)

Applying linear models : Parameter selection

- Determining P and Q.
- Simplest way (if one has multiple time series of the same kind): Select the parameters so as to minimize the RMSE over the validation set.
- More advanced: Look at the autocorrelation function (ACF) and the partial autocorrelation function (PACF)!
- Usage: (explanations on the next slides)

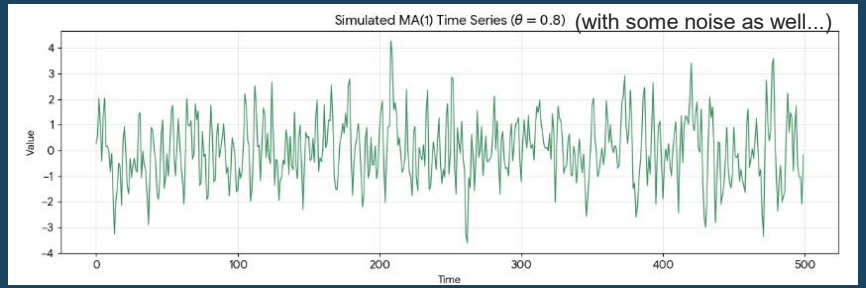
Process	ACF shape	PACF shape
AR(p)	Tails off (~sinusoidal or exponential)	Sharp cutoff after lag p
MA(q)	Sharp cutoff after lag q	Tails off (~sinusoidal or exponential)
ARMA(p,q)	Tails off (~sinusoidal or exponential)	Tails off (~sinusoidal or exponential)

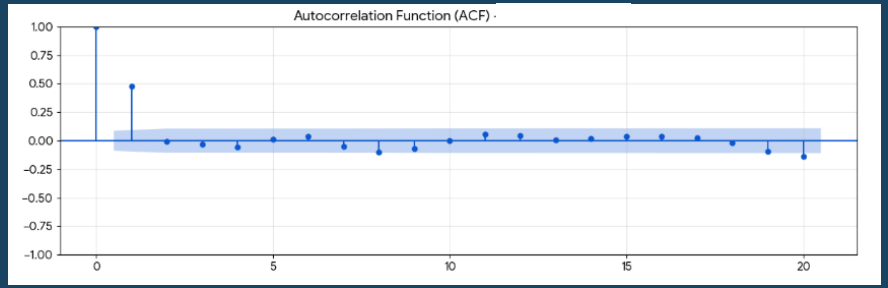
Autocorrelation function (ACF)

- The ACF (in the range [-1,1]) measures the covariance between a time series and its lag, normalized by the variance:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\sum_{t=k+1}^{T}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

- Example:



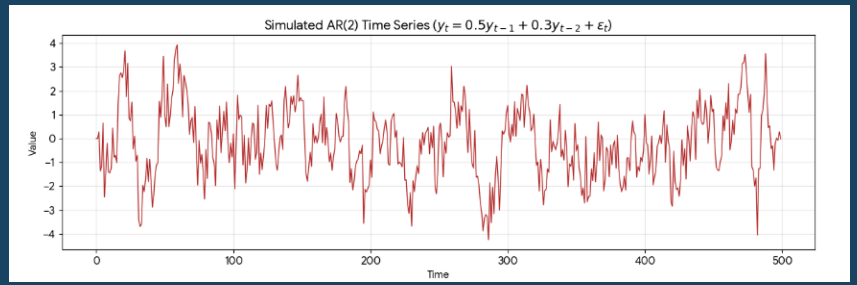


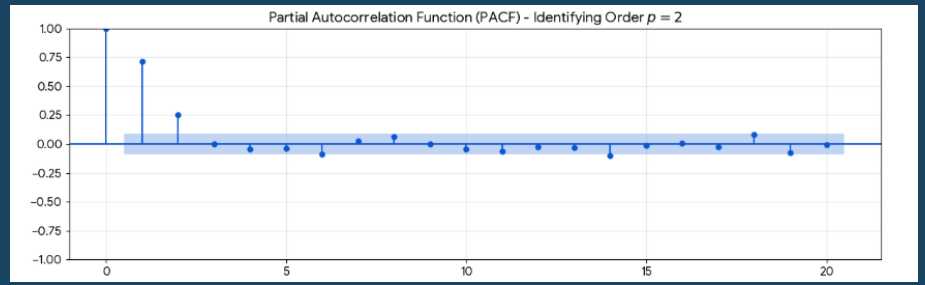
Partial autocorrelation function (PACF)

- The PACF (also in the range [-1,1]) measures the correlation between y_t and y_{t-k} that is *not* explained by the intermediate lags $(y_{t-1}, y_{t-2}, \ldots)$.
- Method: Fit an AR(k) model (see below)

$$y_t = \phi_{k1} y_{t-1} + \phi_{k2} y_{t-2} + \cdots + \phi_{\mathbf{kk}} y_{t-k} + \epsilon_t$$

- The PACF is then the value of ϕ_{kk}.
- Example:





Applying linear models : Fitting

- The final step in linear time series modelling (after selecting p and q for an ARMA model) is to actually find the model coefficients ϕ_i and θ_i.
- There are many methods for doing so.
- For ARMA one usually applies maximum likelihood estimation (MLE), a probabilistic approach that, essentially, asks the question: *What values of ϕ_i and θ_i make the observed data the most likely outcome?*

$$X_t = c + \underbrace{\phi_1 X_{t-1} + \cdots + \phi_p X_{t-p}}_{\text{AR Part}} + \underbrace{\theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}}_{\text{MA Part}} + \epsilon_t$$

MLE parameter fitting for ARMA

- Step 1: Select a model for the noise, usually that the noise terms are independent and identically distributed (i.i.d.) and follow a normal distribution:

$$\epsilon_t \sim N(0, \sigma^2)$$

Probability of observing a speific error:

$$f(\epsilon_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\epsilon_t^2}{2\sigma^2}\right)$$

- Step 2: Likelihood function: Product of the probabilities of all errors.
  - Problem: Small number due to repeated multiplication
  - Solution: Use log-likelihood:

$$\ell(\phi, \theta, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=1}^{n}\epsilon_t^2$$

MLE parameter fitting for ARMA

- Step 3: Compute the errors
  - Problem: we only observe y_t, not e_t
  - Solution: Rearrange the equation, here exemplified for ARMA(1,1):

$$\epsilon_t = y_t - \phi y_{t-1} - \theta \epsilon_{t-1}$$

  - Start with an initial guess (e.g., $\phi = 0.1, \theta = 0.1$), then compute all errors iteratively => $l(\phi, \theta, \sigma^2)$.
  - For y_0 and ϵ_0 one can use the value 0 (but more sophisticated methods exist).
- Step 4 (schematically)
  - Tweak the parameters (ϕ and θ) a little bit (using a suitable method – many different (numerical) methods available, e.g., Nelder-Mead) – apply gradient descent (essentially)
  - Then recalculate the log-likelihood, and so on ...

Python pointers

- AR(I)MA models are easy to implement: You need to write one method (e.g.,. arma_inference) for making predictions and one (e.g., arma_loss) for computing the loss function.
- Thus, you will need to do some coding here, but you can then use the `minimize` command in `scipy.optimize`, where you can specify
  the Nelder-Mead (derivative-free method) that converges nicely:

```
res = minimize(arma_loss, np.zeros(p+q), method='Nelder-Mead', tol=1e-4)
```

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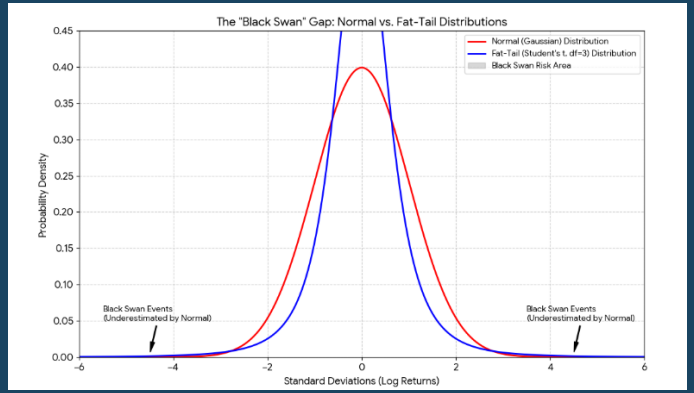
Black swans!

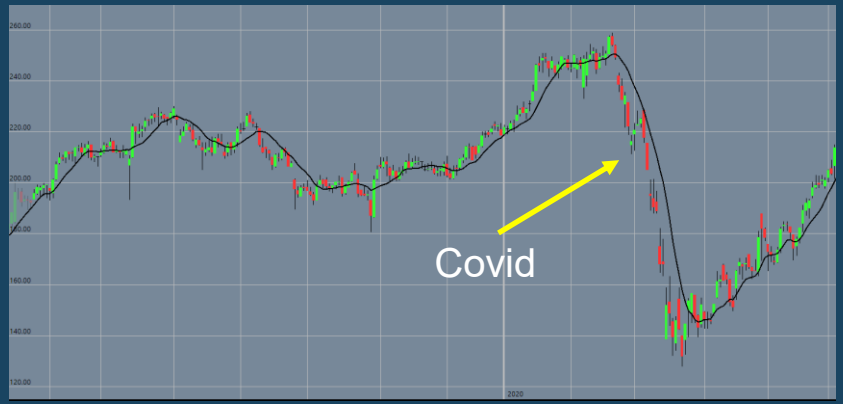
- Many time series exhibit regular patterns over long times.
- However, they may then undergo variations that are ...
  - ... rare
  - ... very strong
  - ... unpredictable (by existing theories)
- This is known as the *black swan phenomenon*.

Black swans: Challenges

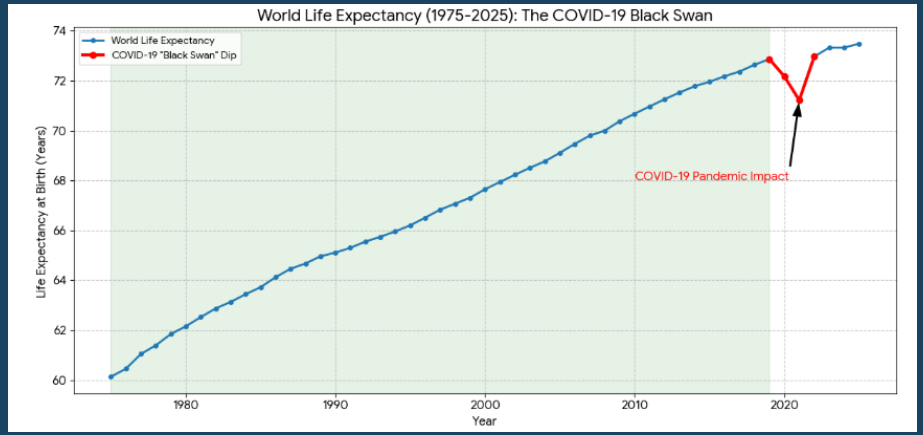
- The assumption of normal distribution
  - Often holds rather well, but some series (e.g., financial time series) exhibit so-called *fat tails* – an excess of rare, high-impact events.
- The induction problem
  - Models are induced from historical data – difficult to account for rare events.
- Overfitting
  - Black swans will be treated as strong noise – not properly modelled.

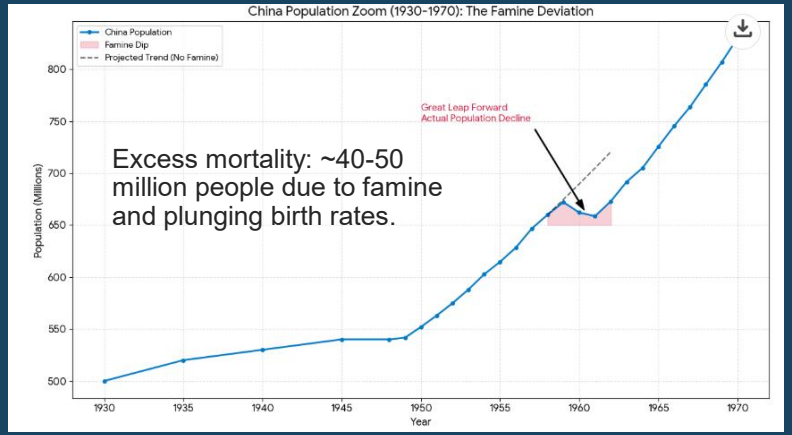


Black swans: Real-world examples (1)



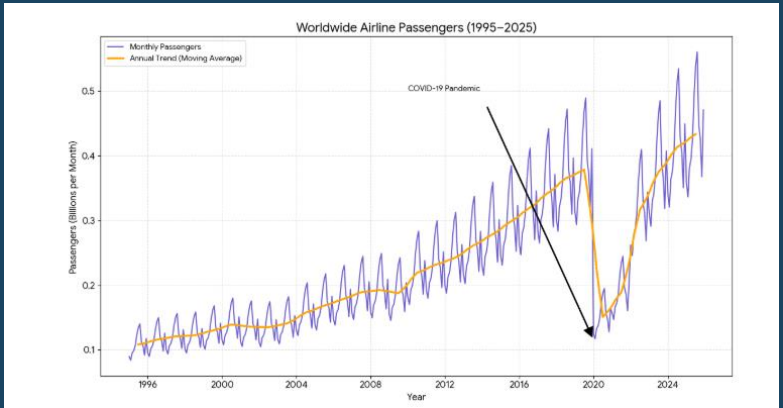
Black swans: Real-world examples (2)

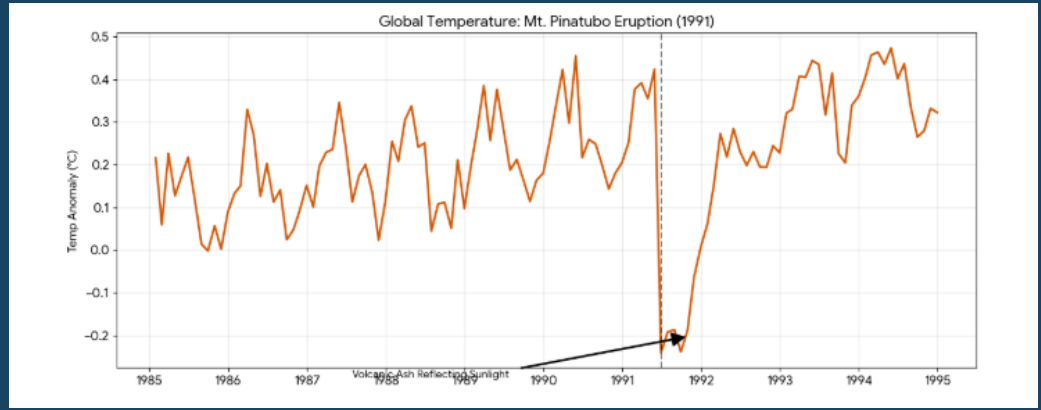




Excess mortality: ~40-50 million people due to famine and plunging birth rates.

Black swans: Real-world examples (3)





Black swans: Final considerations

- One cannot, in general, use ARMA to model time series with massive swings.
- An alternative is to use so-called GARCH (generalized autoregressive conditional heteroskedasticity) models.
- GARCH is designed to handle volatility clustering (periods of high variance)
- Comparison:
  - In ARMA, the variance (volatility) is constant (homoskedasticity)
  - In GARCH, the variance is itself modeled as a time series (heteroskedasticity)
- We will not study GARCH models further here, though.

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Black-box models

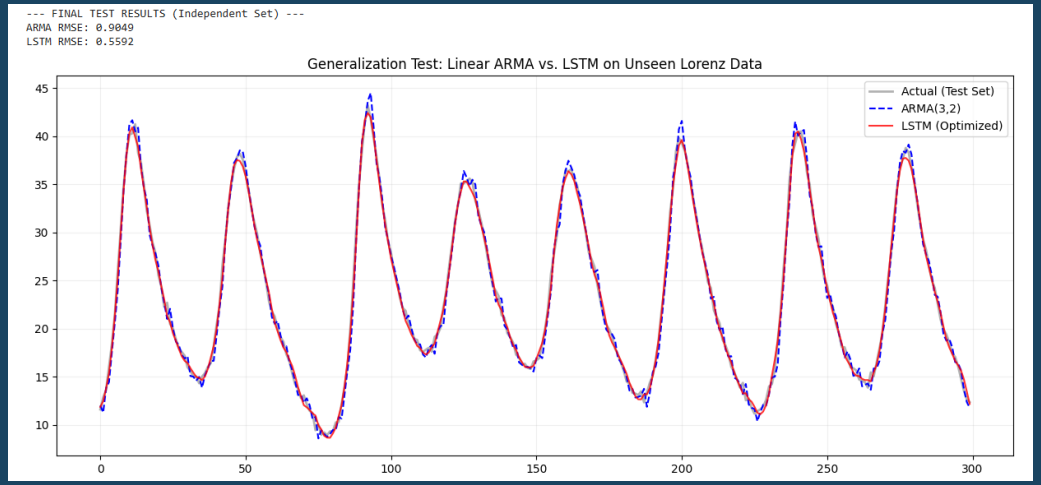
- One can also apply neural networks for time series prediction, for example ...
  - RNNs (standard neurons): Captures short to intermediate-range dependencies, but suffers from vanishing gradients
  - LSTM networks: Captures both short-range and long-range dependencies
  - CNNs: Treats a time series as an image, trying to find patterns.
  - Transformers: Use attention to investigate the impact of past events (possibly very far in the past).
- There are also approaches that mix linear and black-box models (e.g., using an ARIMA part for the trend and a neural network for the residuals).
- Here we will only consider LSTMs.

LSTMs for time series prediction

- Usually a good choice of model, unless one has huge amounts of data.
- For small to medium size data sets (~thousands of data points rather than many millions) LSTMs often outperform transformers (that would normally overfit the data).
- Applies, for instance to most financial data sets
  - Example: 30 years of data x 252 trading days x 100 stocks = 756,000 data points (but also very similar variation in many stocks (same shocks etc.) => effectively a smaller set).
- Typically with one or two LSTM layers (10-100 units each), and an output layer with a single neuron (predicting the next value in the time series)

LSTMs for time series prediction

- Typical result: The (single-layer) LSTM network is a bit better (though not so very much), especially where the series changes rapidly.
- The plot on the right shows an ARMA model (blue, dashed) vs. an LSTM predictor (red), trained with holdout validation and then applied to a previously unseen data set:
- Note that all prediction are single-step.



Python pointers

- LSTMs are available in Tensorflow Keras (`tensorflow.keras.models` and `tensorflow.keras.layers`).
- Setting up a single-layer LSTM (”Sequential” comes from tensorflow.keras.models):

```
lstm = Sequential([
    LSTM(32, activation='tanh', input_shape=(window_size, 1)),
    Dense(1)
])
lstm.compile(optimizer='adam', loss='mse')
```

- window_size = the length of the lookback. Per default, the LSTM is ”stateless” – the entire hidden state is wiped after each window has been processed.
- Early stopping (holdout validation) should be applied when training the LSTM network

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Predicting events

- In some cases one aims to predict (rather) rare (but recurrent) events, where a time series makes a bigger move than usual.
- Note: This is not the same as trying to predict black swans!
- Example: Stock market prediction:
  - Most days are not relevant to predict.
  - Trying to do so also risks fitting the model to the noise.
  - Instead, make predictions only for the days marked with arrows.
  - How can one find those days?
  - (See the next slides!)

Predicting events

- This approach effectively turns the prediction problem into a *classification* problem.
- For example, in the image shown on the previous page, the desired output would be 0 for all time steps, except those with arrows, where it would be 1.
- Aim: *Classify* preceding patterns (time series variation) such that...
  - ... the output is 0 for most time steps, but
  - ... 1 for the selected time steps.

Predicting events: Approaches

- Assuming that the data sets (training, validation, and test) have been prepared as illustrated on the previous slide, several different approaches can be used, e.g.,
  - LSTM networks:
    - Train the network (with holdout validation) to (try to) predict 0 for all events (days) except the selected ones, where it should output 1.
    - When running the network, it will give outputs y in the range [0,1]. Thus, select a threshold T (in that range).
    - If $y > T$, then output a 1, otherwise 0.
    - The validation set can be used for finding a suitable value of T.
  - Evolvable rule-based systems:
    - Evolve (structurally and parametrically) a set of rules that aims to output a 0 for all events (days) except the selected ones, where it should output a 1.
    - Use holdout validation (as above).
    - When running the system, just use its raw output (either 0 or 1).
    - This kind of system will be covered in a later lecture, and is relevant for Assignment 6 (voluntary).

Today’s learning goals

- After this lecture you should be able to
  - Describe and discuss different types of time series ✓
  - Describe and compare different linear models ✓
  - Apply linear models to a time series prediction task ✓
  - Describe and discuss the Black swan phenomenon ✓
  - Apply LSTM networks to a time series prediction task ✓
  - Briefly describe event prediction (as a classification task) ✓

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